

Predicting and Characterising Air Pollution Patterns in Beijing Using Machine Learning

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Abstract

This study investigates the application of machine learning techniques to predict and characterise air pollution patterns in Beijing, using the Beijing Multi-Site Air Quality Dataset covering the period from March 2013 to February 2017. The analysis combines supervised and unsupervised learning approaches applied to hourly measurements of six pollutants and six meteorological variables collected across twelve monitoring stations.

On the supervised learning side, three model families — Ordinary Least Squares, Ridge regression, and Histogram Gradient Boosting — were evaluated across three feature sets of increasing richness. The results show that feature engineering, particularly the inclusion of recent PM_{2.5} lag and rolling mean features, contributed far more to predictive performance than model complexity. The final model, Histogram Gradient Boosting with the full feature set, achieved a test RMSE of 17.04 $\mu\text{g}/\text{m}^3$ and a test R^2 of 0.958, explaining approximately 96% of the variance in held-out hourly PM_{2.5} concentrations. Permutation feature importance confirmed that the 1-hour PM_{2.5} lag is by far the dominant predictor, followed by co-pollutants such as CO and NO₂.

On the unsupervised side, K-Means clustering combined with Principal Component Analysis identified four physically interpretable pollution regimes: Clean/windy days, Moderate conditions, Summer photochemical haze, and Severe pollution episodes. The PCA analysis revealed that pollution intensity, driven by combustion-related pollutants, is the dominant axis of variability in the data, while a secondary axis captures abnormal meteorological warmth associated with summer conditions.

Both approaches converged on the same key findings: recent PM_{2.5} history, wind speed, CO and NO₂ are the most important variables driving air quality variations in Beijing, and seasonal and meteorological conditions play a central role in shaping pollution levels.

Keywords: PM_{2.5} prediction, air quality, machine learning, gradient boosting, K-Means clustering, PCA, Beijing

1 Introduction

Air pollution represents a critical environmental and public health issue, particularly in densely populated urban areas where emissions and meteorological conditions interact in complex ways. The increasing availability of environmental data has created new opportunities to better understand, predict, and manage air quality through data-driven approaches. In this context, machine learning techniques provide powerful tools for both predictive modeling and pattern discovery. We selected air pollution as a topic because it is directly related to sustainability and environmental management, and because the available dataset was sufficiently rich for both predictive modelling and pattern discovery.

The objective of this study is to investigate whether machine learning models that we saw in class can effectively predict air pollution levels and identify underlying patterns across different locations and time periods. More specifically, this work aims to develop predictive models capable of estimating concentrations of key pollutants, such as fine particulate matter (PM_{2.5}), based on temporal, meteorological, and spatial variables. The project focuses not only on supervised learning, but also on unsupervised learning — the study also seeks to identify recurring pollution regimes through unsupervised clustering.

To achieve these objectives, we applied several machine learning approaches. For supervised learning, we evaluated OLS, Ridge regression, and Histogram Gradient Boosting to predict continuous hourly PM_{2.5} concentrations. In parallel, we explored unsupervised learning techniques to uncover hidden structures in the data, including Principal Component Analysis (PCA) for dimensionality reduction and K-Means clustering to identify recurring pollution regimes across time and monitoring stations.

Our analysis is conducted using the Beijing Multi-Site Air Quality Dataset, which provides hourly measurements of six air pollutants, including PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃, along with meteorological variables and timestamps collected from twelve monitoring stations between 2013 and 2017. This dataset offers a detailed view of both temporal dynamics and spatial variability in air pollution, which made it well suited to our study.

By combining predictive modeling and exploratory data analysis, this study aims not only to forecast pollution levels but also to understand the key factors and patterns that drive air quality variations in urban environments.

2 Literature Review

Air pollution, and fine particulate matter (PM_{2.5}) in particular, represents a major public health challenge in urban environments, with well-documented links to respiratory and cardiovascular disease and premature mortality (World Health Organization, 2018). Traditional monitoring networks offer limited spatial and temporal coverage, motivating a growing body of research into machine learning as a complementary approach for air quality prediction and pattern discovery.

On the supervised learning side, gradient boosting methods have demonstrated strong performance for tabular prediction tasks. Chen and Guestrin (2016) introduced XGBoost, a scalable tree boosting system that has since become one of the most widely used ensemble methods for regression problems on structured data. While our study uses Histogram Gradient Boosting rather than XGBoost specifically, both methods share the same core principle of sequentially building trees to correct prediction errors. The effectiveness of such approaches for air quality prediction has been demonstrated in contexts directly relevant to our study: Lyu et al. (2022) applied multiple machine learning algorithms to predict spatiotemporal variations of air pollutants including PM_{2.5} and ozone in the Beijing-Tianjin-Hebei region from 2014 to 2021, finding that machine learning models effectively captured the complex relationships between meteorological conditions and pollutant concentrations. Meteorological variables are consistently identified as critical features in predictive models of air quality. Guo et al. (2023) demonstrated that meteorological elements such as temperature, wind speed and humidity are key drivers of daily PM_{2.5} concentrations in urban environments, motivating the inclusion of these variables as primary features in our supervised models.

On the unsupervised side, PCA and clustering have been widely applied to uncover latent structures in air quality data. Pires et al. (2008) demonstrated that combining principal component analysis and cluster analysis is effective for managing air quality monitoring data across multiple stations, showing that PCA successfully identifies the dominant sources of pollutant variability while clustering reveals distinct pollution microenvironments. More recently, Elmourssi et al. (2026) applied a combined framework of K-Means clustering and machine learning classifiers to identify four distinct air pollution regimes in Greater Cairo, showing that unsupervised clustering can successfully characterise recurring pollution conditions even in complex urban environments. This approach directly mirrors our own unsupervised methodology, where K-Means clustering is used to identify four recurring pollution regimes across the twelve Beijing monitoring stations, with PCA used for dimensionality reduction

and visualisation.

The Beijing Multi-Site Air Quality dataset used in this study is well established in the literature, providing hourly measurements of six pollutants and six meteorological variables across twelve nationally-controlled monitoring sites from March 2013 to February 2017 (Zhang et al., 2017). The present study contributes to this body of work by applying an integrated framework combining regression and unsupervised learning within a single analysis, with an emphasis on interpretable models and pattern discovery rather than predictive performance alone.

3 Methodology

3.1 Data Description

The dataset contains hourly measurements collected from 12 monitoring stations distributed across Beijing, covering the period from March 2013 to February 2017. The variables can be grouped into two main categories: pollutant measurements and meteorological variables.

Pollutant variables:

- **PM_{2.5}:** Fine particulate matter smaller than 2.5 micrometers in diameter. It is the target variable of this study and the primary variable of interest throughout the analysis. We use it as the main variable related to pollution because it can be particularly harmful to human health as these particles can penetrate deep into the lungs (Lelieveld et al., 2015).
- **PM₁₀:** Slightly larger particles (up to 10 micrometers). It shares similar emission sources with PM_{2.5} (traffic, industrial activity, heating).
- **NO₂:** Nitrogen dioxide, it is primarily emitted by vehicles and heating systems. As we will see through the EDA, its concentration tends to rise alongside PM_{2.5} during pollution episodes.
- **CO:** Carbon monoxide, a combustion-related pollutant primarily emitted by vehicles, heating systems, and industrial activity.
- **SO₂:** Sulfur dioxide, mainly linked to coal burning and industrial activity. It follows a similar seasonal pattern as PM_{2.5} and is most prevalent during winter months.
- **O₃:** Ozone, which behaves differently from the other pollutants. Rather than being directly emitted, it forms through chemical reactions related to sunlight.

Meteorological variables:

- **WSPM:** Wind speed, it is arguably the most impactful meteorological variable in this study. Higher wind speeds are consistently associated with lower PM_{2.5} concentrations, as wind physically disperses pollutants. We will further develop this relationship in the EDA.
- **TEMP:** Temperature of the atmosphere, measured in degrees Celsius. We will see in the unsupervised analysis that we use monthly temperature anomalies rather than raw values, in order to detect days that

are unusually warm relative to the seasonal norm.

- **DEWP:** Dew point, it is a measure of atmospheric humidity. Like temperature, it is used in its anomaly form in order to detect days that are unusually humid relative to the seasonal norm.

3.2 Data Preprocessing

Prior to applying the machine learning models we want to use, a crucial step was to prepare the dataset to ensure its quality and suitability for the analysis. Although the Beijing Multi-Site Air Quality Dataset we used is relatively well-structured and already quite clean, it still contains a number of missing values, which can be common in environmental monitoring systems due to numerous potential reasons, such as sensor malfunctions or irregular data collection.

To address this issue, we thought that implementing a temporal interpolation strategy was the best option. Temporal interpolation is a method used in time-series analysis to estimate missing observations based on the values recorded at neighboring time points. We thought that it was a better option than discarding incomplete records, which we believed could result in a significant loss of information.

We believe that in the context of the data that we had, air pollution data, temporal interpolation was appropriate due to the inherent continuity of environmental processes and because of the quite big dataset we had. Indeed, we know that pollution levels tend to evolve progressively over time, making it reasonable to approximate missing values using adjacent observations. This approach preserves the temporal structure of the data and ensures compatibility with the machine learning algorithms we planned to use, which generally require complete input datasets.

We applied linear temporal interpolation with a maximum gap limit of 6 consecutive hours. This means that for any sequence of missing observations, if the gap did not exceed 6 hours, the missing values were estimated by linearly interpolating between the last known value before the gap and the first known value after it. Gaps exceeding this threshold were instead filled using station-level median imputation, as interpolating over longer periods would risk bridging across distinct atmospheric regimes, such as nighttime stagnation and daytime convective mixing, where a linear assumption becomes unreliable.

Because data cleaning was performed before the temporal train/test split, we explicitly verified that the lag and rolling features do not use future PM2.5 values: they are computed within each station using only past observations. The main potential source of information leakage would be imputation across the train/test boundary. However, the interpolation limit was restricted to short gaps of at most six hours, and longer gaps were handled with station-level median imputation. This makes any possible boundary effect highly local and limited, but it remains a methodological limitation of the preprocessing pipeline.

3.3 Supervised Learning

The supervised part of this study addresses a regression problem: predicting hourly PM2.5 concentrations at the twelve Beijing monitoring stations. A strict temporal train/test split was used, with the first 80% of observations used for training and the remaining 20% reserved for testing. This ensures that the model is always evaluated on future observations rather than on randomly mixed data, which would introduce data leakage and overestimate real-world performance.

3.3.1 Feature Engineering

Rather than using the raw variables directly, we constructed three feature sets to systematically assess the contribution of different types of information. Feature set A contains only weather variables (temperature, dew point, wind speed, pressure, precipitation), temporal variables (cyclic hour and month encodings, season), wind direction encodings, and station dummies. Feature set B contains only PM2.5 lag and rolling mean features, specifically the 1-hour, 3-hour, and 6-hour lags, as well as the 3-hour, 6-hour, and 24-hour rolling means. Feature set C combines the contextual variables from set A, the PM2.5 lag and rolling mean features from set B, and co-pollutants (CO, NO₂, SO₂, O₃), while deliberately excluding PM10 to avoid redundancy with PM2.5 given their very high correlation. Although a log-transformation of PM2.5 was considered because of the right-skewed target distribution, the final supervised models were trained directly on raw PM2.5 concentrations. This choice produced better predictive performance and keeps RMSE and MAE directly interpretable in $\mu\text{g}/\text{m}^3$.

3.3.2 Models

Three model families were evaluated across all three feature sets. Ordinary Least Squares (OLS) serves as the linear baseline. Ridge regression adds L2 regularisation to OLS, penalising large coefficients to reduce potential overfitting. Histogram Gradient Boosting (HGB) is a non-linear ensemble method that builds trees sequentially, with each tree correcting the errors of the previous one. HGB is a computationally efficient variant of Gradient Boosting that uses histogram-based binning of continuous features, making it well-suited for large datasets such as ours. All models are evaluated using RMSE, MAE and R^2 on the held-out test set.

3.3.3 Interpretation

To understand which features drive the model's predictions, we computed permutation feature importance on the test set, measuring the increase in RMSE when each feature is randomly permuted. We also produced Partial Dependence Plots (PDPs) for the five most important features, showing the marginal association between each feature and the predicted PM2.5 while averaging over the distribution of all other features.

3.4 Unsupervised Learning

The unsupervised part of this study aims to identify natural groupings of pollution patterns across the 12

Beijing monitoring stations, using a combination of K-Means clustering and Principal Component Analysis (PCA).

3.4.1 Data Construction

For the unsupervised analysis, the data were aggregated to daily station-level means, so that each observation represents one station on one day. This aggregation also removes the hourly noise while preserving the overall pollution profile of each day. Nine variables were retained for our clustering: six pollutants (PM2.5, PM10, SO₂, NO₂, CO, O₃) and three meteorological variables (temperature, dew point, wind speed), selected based on exploratory data analysis and feature importance results from the supervised analysis.

3.4.2 Preprocessing

Three issues were identified in the raw features and corrected prior to modelling. First, the four most skewed pollutants (PM2.5, PM10, CO, SO₂) were log-transformed using a log1p transformation to reduce the influence of extreme concentration events on distance calculations. Second, temperature and dew point were replaced by their monthly station-level anomalies, which is the difference between the observed value and the historical monthly mean for that station, in order to capture abnormal meteorological conditions rather than the regular seasonal cycle. Third, all variables were winsorised at the 1st and 99th percentiles to limit the impact of outliers without discarding those observations. A StandardScaler was finally applied to bring all features onto a common scale, which is a necessary condition for both K-Means and PCA to function correctly.

Dimensionality reduction — PCA. Principal Component Analysis is a linear dimensionality reduction technique that transforms the original variables into a new set of uncorrelated axes, called principal components, that is then ordered by the amount of variance each PC explains. Each principal component is a combination of the original feature, and the first two components together capture the directions of greatest variability in the data. In this study, PCA serves two purposes: it allows us to understand which variables drive the most variance through the examination of component loadings, and it enables a two-dimensional visualisation of the clusters, since the original nine-dimensional space cannot be represented graphically. It is important to note that clustering itself was performed in the full nine-variable space — PCA is used for interpretation and visualisation only.

Clustering — K-Means. We used K-Means as our clustering algorithm. It partitions observations into k groups by iteratively assigning each point to its nearest center and updating centers as the mean of their assigned points, until convergence. The goal of the algorithm is to minimize the total within-cluster inertia, defined as the sum of squared distances between each point and the center of the cluster. The number of clusters k is a hyperparameter that must be selected before fitting the model. To select an appropriate value, we used two

complementary diagnostics: the elbow method, which plots inertia against k and looks for a point of diminishing returns, and the silhouette score, which measures how well each observation fits its own cluster relative to neighbouring ones, ranging from -1 to $+1$. The results of these diagnostics are discussed in the Results section, where we justify why we chose $k = 4$ based on both the quantitative criteria and the interpretability of the clusters.

4 Results

4.1 Exploratory Data Analysis

Before jumping into the final results we made with the models, we did an exploratory data analysis to familiarize ourselves better with the data we were going to use and its different patterns. In this sub-section we will shortly go through some main graphs and aspects we found during this part of the project.

4.1.1 PM2.5 Distribution

The first step of the analysis consists in examining the distribution of PM2.5, which is the target variable of this study. As shown in Figure 1, we can see that the raw distribution is right-skewed, with a mean of $79.7 \mu\text{g}/\text{m}^3$ significantly exceeding the median of $55.0 \mu\text{g}/\text{m}^3$, a sign of a log-normal distribution driven by some rare extreme pollution episodes. There are some values extending beyond almost $500 \mu\text{g}/\text{m}^3$, which corresponds to severe pollution events observed during Beijing winters.

It seems important to state here that this distributional asymmetry was considered during the modelling process. For supervised learning, a $\log(1 + \text{PM2.5})$ transformation of the target variable was tested, but it did not improve the final predictive performance. Therefore, the final supervised models were trained directly on raw PM2.5 concentrations, which keeps RMSE and MAE directly interpretable in $\mu\text{g}/\text{m}^3$. For unsupervised learning, however, the log-transformation was retained for the four most skewed pollutants before clustering, because K-Means minimises Euclidean distances and can therefore be sensitive to extreme values.

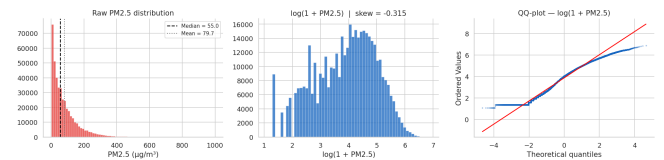


Figure 1. Distribution and spread of PM2.5.

After applying the log transformation for exploratory purposes, the skewness drops to -0.315 and the distribution becomes approximately symmetric. The QQ-plot in Figure 1 confirms this improvement in the central range, while the slight deviations in both tails reflect residual extreme pollution episodes. These deviations do not compromise the usefulness of the transformation for exploratory analysis and for the unsupervised clustering preprocessing, where reducing skewness helps limit the influence of extreme values.

4.1.2 Seasonal Patterns

We can see in Figure 2 a seasonal pattern in the PM2.5 concentrations across the four different seasons. Indeed, winter stands out as the most polluted season, with the widest interquartile range and the also the highest concentration of extreme values. We see that several observations exceed $800 \mu\text{g}/\text{m}^3$, corresponding to big pollution episodes. Summer shows the lowest pollution levels, likely due to higher winds and rain which naturally clean the air. Indeed, Beijing has a monsoon climate, where summer precipitation accounts for more than 70% of annual rainfall (Yang et al., 2020). Spring and Autumn occupy intermediate positions, with Autumn showing a slightly higher spread reflecting the early onset of the heating season.

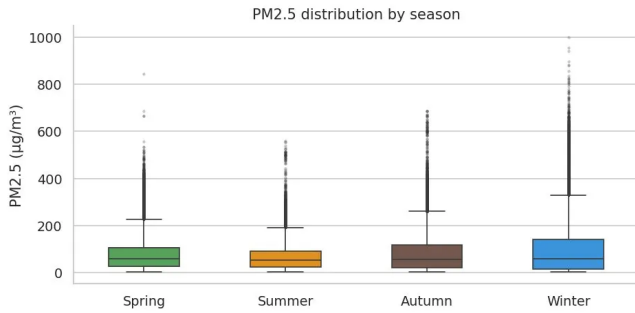


Figure 2. PM2.5 distribution by season.

This seasonal pattern creates a problem that we will further see in the clustering analysis. Since temperature naturally follows a similar cycle as pollution, cold in winter, warm in summer, a clustering algorithm could simply group days by season rather than by actual pollution conditions. This means that, without any correction, the clusters might just tell us “it is winter” or “it is summer”, which is not very useful. We will see later how we addressed this problem in the unsupervised part of the report.

4.1.3 Temporal Patterns

Figure 3 reinforces this point. The heatmap displays the median PM2.5 for every combination of month and hour of the day. We can see two patterns that stand out, confirming the first impressions we had with our EDA. First, the seasonal contrast is immediately visible: months 1, 2, 3, 11 and 12, which are the winter months, are almost systematically “redder” confirming that pollution is structurally higher regardless of the time of day. The only exception is the February month. It might be related to the Lunar New Year holiday, which falls in January or February and causes a significant temporary reduction in industrial activity, traffic, and energy consumption across Beijing, which might explain the measurably lower PM2.5 concentrations despite being in the winter season. Summer months (6, 7, 8) are almost uniformly light (a bit less for July), with median values rarely exceeding $50 \mu\text{g}/\text{m}^3$ at any hour.

Figure 4 adds a third dimension to the temporal analysis by crossing the hour of day with the day of the week. The daily cycle that we just observed; higher pollution at night and in the early morning, lower in the afternoon is

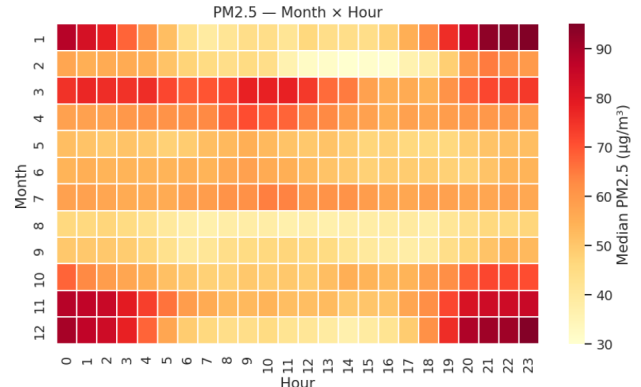


Figure 3. Heatmap of PM2.5 by month and hour.

consistent across all seven days. We can suggest that it is driven by atmospheric physics rather than by human activity. However, a notable difference emerges at the end of the week: Saturday and Sunday nights (22h–2h) show noticeably “redder” cells, suggesting higher pollution during weekend late-night hours. This could reflect increased traffic and social activity later in the evening compared to weekdays. Monday mornings also stand out as relatively clean, possibly due to lower industrial activity at the start of the week, which is followed by a little weekend slowdown. Overall though, the day-of-week effect remains modest compared to the seasonal and hourly effects identified earlier. It does explain less variance in PM2.5 and we’ll therefore treat it as a secondary factor in the modelling.

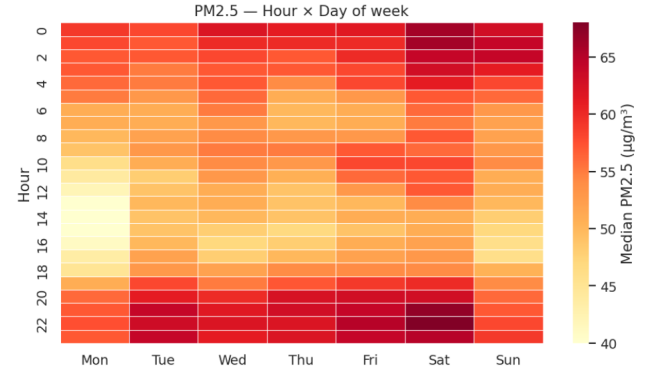


Figure 4. Heatmap of PM2.5 by hour and days of the week.

4.1.4 Correlation Analysis

We wanted to have more information on the importance of the variables when predicting the PM2.5, that is why we did a Pearson correlation matrix across all numeric features. Several key patterns emerge.

The first thing that emerges is the dominance of lag and rolling features. The 1-hour lag (pm25_lag_1h) reaches a correlation of 0.97 with PM2.5, which almost represents a perfect linear relationship, meaning that the pollution level of the previous hour is very predictive of the next hour’s pollution level. The 3-hour lag (0.88) and the rolling averages over 3h (0.94) and 6h (0.90) remain extremely high. This reflects the strong temporal persistence of pollution: once a pollution episode starts, it tends to last for several hours. This result is non-trivial, it means that for short-term prediction, knowing

recent past values is more informative than knowing any meteorological or chemical variable. It directly justifies their inclusion as primary features in the supervised models.

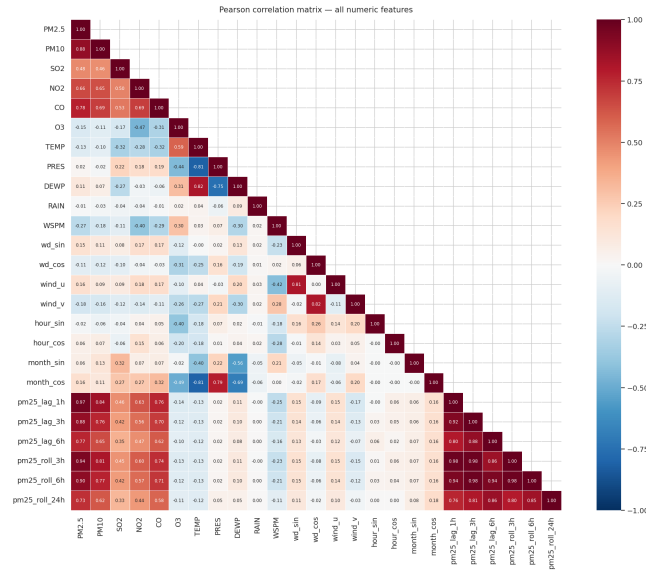


Figure 5. Pearson correlation matrix.

Among physical variables, combustion-related pollutants are the strongest predictors: PM10 ($r = 0.88$), CO ($r = 0.78$) and NO₂ ($r = 0.66$) all show strong positive correlations, consistent with their common emission sources in traffic and heating. Wind speed (WSPM, $r = -0.27$) and temperature (TEMP, $r = -0.13$) show negative correlations — The most plausible explanation for these values is that higher wind speed disperses particles, while warmer conditions are associated with cleaner air, as already briefly discussed in the seasonal analysis.

For the clustering analysis, where we will see that we did deliberately exclude the lag features since the goal is more to characterize pollution regimes rather than predict future values, this matrix confirms that PM10, CO, NO₂ and WSPM are the most informative variables to retain.

4.1.5 Wind Speed Effect

Figure 6 illustrates this negative relationship more concretely by breaking wind speed into five categories. As we can see, the pattern is very clear: as wind speed increases, both the median PM2.5 and the spread of the distribution drop consistently. At very low wind speeds (0–1 m/s), the median sits around $90 \mu\text{g}/\text{m}^3$ with extreme values reaching $1000 \mu\text{g}/\text{m}^3$. At speeds above 5 m/s, the median falls below $20 \mu\text{g}/\text{m}^3$ and the interquartile range becomes very narrow. This confirms that wind is not just correlated with lower pollution — it acts as a physical dispersal mechanism that actively limits the accumulation of particulate matter in the atmosphere. Days with near-zero wind are precisely the conditions under which severe pollution episodes develop and persist, which directly connects to some of the clusters we will later see in the unsupervised analysis.

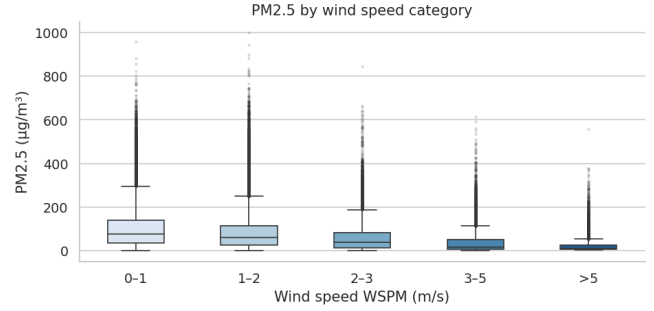


Figure 6. PM2.5 by wind speed category.

4.2 Supervised Learning Analysis

4.2.1 Feature Sets and Models Tested

Before selecting a final model, we wanted to evaluate three model families, OLS, Ridge regression, and Histogram Gradient Boosting (HGB), across three feature sets of increasing richness.

The three feature sets are the following: feature set A contains only weather, temporal, wind direction, and station variables. Feature set B contains only PM2.5 lag and rolling mean features. Feature set C combines the contextual variables from set A, the lag and rolling features from set B, and co-pollutants (CO, NO₂, SO₂, O₃), while excluding PM10 to avoid redundancy with PM2.5.

Table 1. Full model comparison table.

Model	Set	Tr.RMSE	Te.RMSE	Tr. R^2	Te. R^2
HGB	C	15.04	17.04	0.965	0.958
OLS	C	18.72	17.41	0.946	0.956
Ridge	C	18.72	17.41	0.946	0.956
Ridge	B	20.18	18.77	0.937	0.949
OLS	B	20.18	18.77	0.937	0.949
HGB	B	19.80	19.38	0.939	0.945
HGB	A	49.13	56.63	0.625	0.534
OLS	A	63.29	65.19	0.377	0.382
Ridge	A	63.29	65.20	0.377	0.382

The results are summarised in Table 1. The most striking finding is the large performance gap between feature set A and feature sets B and C. Across all model families, replacing the contextual-only feature set A with the lag/rolling feature set B reduces the test RMSE from around $56\text{--}65 \mu\text{g}/\text{m}^3$ down to around $18\text{--}19 \mu\text{g}/\text{m}^3$, a drop of more than 70%. This confirms that recent PM2.5 history is by far the most informative signal for short-term prediction, as already suggested by the correlation analysis in the EDA. Feature set C provides a further incremental improvement, most visible for HGB, which reaches a test RMSE of $17.04 \mu\text{g}/\text{m}^3$ and a test R^2 of 0.958. Since set C adds both the contextual variables from set A and the co-pollutants to the lag/rolling features in set B, this comparison should not be interpreted as isolating the effect of co-pollutants alone.

It is also worth noting that OLS and Ridge produce exactly identical results on feature sets B and C. This is expected behaviour: with over 336,000 training observations and already highly informative features, the regularisation term in Ridge becomes negligible relative to the data, and both models effectively solve the same

equations. Feature set A is the only case where Ridge differs slightly from OLS, confirming that regularisation only helps when the features are weaker.

The slightly lower test RMSE than training RMSE for OLS and Ridge does not indicate leakage or overfitting. It is more likely explained by the temporal composition of the split: the training period covers a broader range of pollution regimes, including several full winter seasons with more extreme episodes, whereas the test period is shorter and appears slightly easier to predict. This is plausible in air-quality time series, where prediction difficulty varies strongly across seasons and years.

Based on these results, HGB with feature set C was selected as the final supervised model, as it achieves the best test performance across all metrics. We can confirm this choice with Figure 7, showing the difference in the test RMSE between the models, and that HGB-C has the lowest result, followed closely by OLS-C and Ridge-C, which confirms the importance of adding all the variables except PM10.

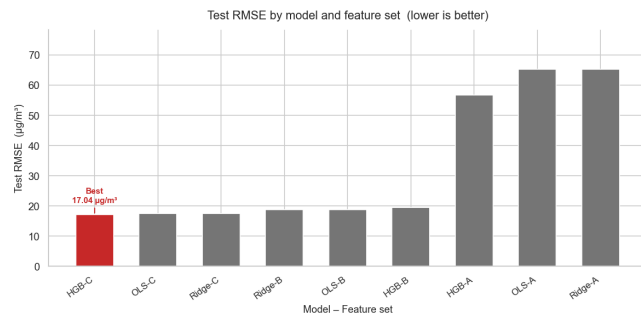


Figure 7. Test RMSE by model and feature set.

4.2.2 Feature Importance

Figure 8 shows the permutation feature importance of the final HGB-C model, measured as the mean increase in RMSE on the test set when each feature is randomly permuted. The results confirm and reinforce the findings we already stated in the model comparison and that we already started to state in the EDA.

Indeed, the 1-hour PM2.5 lag (pm25_lag_1h) is by far the most important feature, with a mean RMSE increase of nearly $60 \mu\text{g}/\text{m}^3$ when permuted. This means that removing access to the most recent PM2.5 value almost completely destroys the model's predictive ability. At one point during our analysis, we considered whether a model without pm25_lag_1h could still be useful, since a model that essentially predicts the next hour from the current hour has limited practical value for longer-horizon forecasting. However, the feature importance results make clear that excluding this feature leads to a heavy loss in performance, and that the model relies on it as its primary anchor. This is an honest limitation of the approach: the model is very good at short-term prediction precisely because pollution changes slowly from one hour to the next, but it would not generalise well to a setting where the most recent observation is unavailable. The 3-hour rolling mean (pm25_roll_3h) and the 3-hour lag (pm25_lag_3h) also contribute, but

their importance is much smaller.

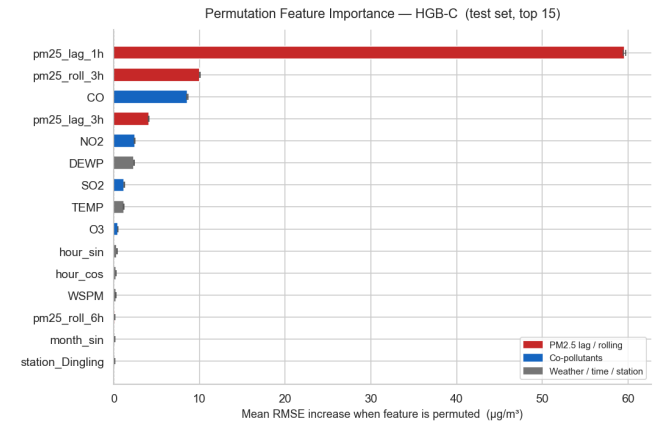


Figure 8. Permutation Feature Importance — HGB-C (test set, top 15).

Among the non-PM2.5 features, CO and NO₂ are the most informative co-pollutants, consistent with their strong correlation with PM2.5 observed in the EDA. SO₂ and O₃ also appear in the top 15 but with smaller contributions. This confirms that adding co-pollutants in feature set C provides genuine additional predictive information beyond the PM2.5 lag features alone.

Finally, weather and temporal variables such as DEWP, TEMP, WSPM, and the cyclic hour features appear at the bottom of the ranking with very small importance values. Their effect is likely partly absorbed by the lag features, which already summarise the recent atmospheric state. Station effects are also minimal, with only one station dummy (Dingling) appearing near the bottom, suggesting that spatial differences between monitoring stations play a secondary role once recent pollution history and co-pollutants are accounted for.

4.2.3 Prediction Diagnostics

Figures 9 and 10 provide two complementary views of the prediction quality of the final HGB-C model on the test set.

The time series plot (Figure 9) shows the first 1,000 test observations, comparing actual and predicted PM2.5 values. The predicted values follow the actual values closely across the majority of the period, capturing both the general level and the short-term fluctuations. This confirms that the model successfully learns the dominant temporal dynamics of pollution. However, some discrepancies are visible at certain peaks, where the predicted line does not fully reach the actual spike.

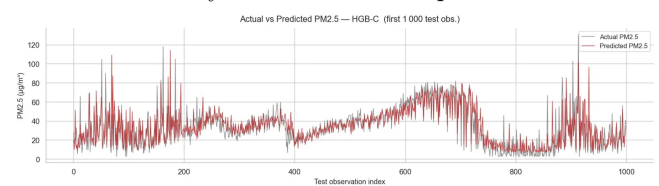


Figure 9. Actual (grey) vs Predicted (red) PM2.5 — HGB-C.

The predicted versus actual scatter plot (Figure 10) confirms this picture on the full test set. The large majority

of points are concentrated along the $y = x$ diagonal, indicating good predictive accuracy across the most common PM2.5 values. For low to moderate pollution levels, the alignment is very tight. However, as actual PM2.5 increases beyond around $400 \mu\text{g}/\text{m}^3$, the scatter becomes noticeably wider and several points fall below the diagonal, meaning that the model tends to underpredict the most extreme pollution episodes. These severe events are rarer in the training data and may depend on sudden local conditions that are not fully captured by the available features. Overall, the diagnostics confirm that HGB-C is a reliable model for general PM2.5 forecasting, while predictions during rare extreme pollution spikes should be interpreted with more caution.

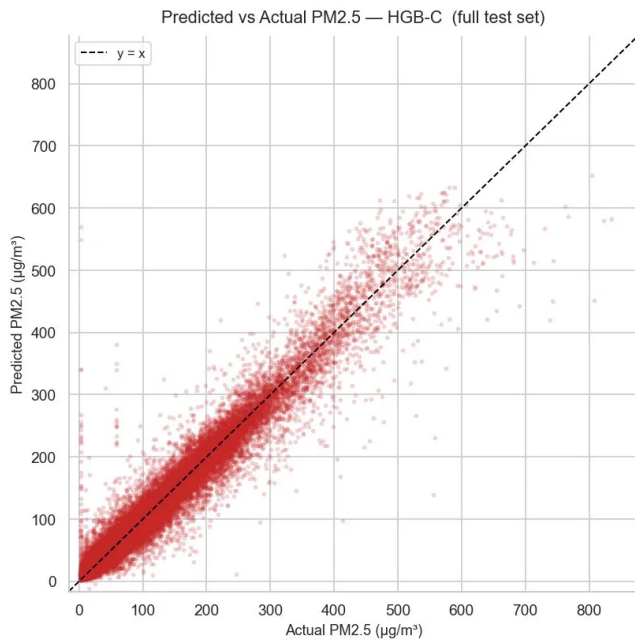


Figure 10. Predicted vs Actual PM2.5 — HGB-C (full test set).

4.2.4 Partial Dependence Plots

Figure 11 shows the Partial Dependence Plots (PDPs) for the five most important features of the HGB-C model, computed on a sample of 5,000 training observations. PDPs show how the model's prediction changes when one feature varies while averaging over the observed distribution of all other features. They should be interpreted as model-learned associations, not causal effects.

The PDP for `pm25_lag_1h` shows a clear and almost perfectly linear positive relationship with predicted PM2.5: as the previous hour's concentration increases, the model predicts a proportionally higher current value. This is the cleanest and most interpretable signal in the model, consistent with everything observed so far.

The PDP for `pm25_roll_3h` shows a more complex and apparently negative pattern. This should not be interpreted as high recent PM2.5 history being protective — it is a statistical artefact caused by the very high correlation between `pm25_lag_1h` and `pm25_roll_3h`. Once the model already knows the most recent hourly value through `pm25_lag_1h`, the rolling mean captures

whether pollution is rising or falling relative to that recent value, which explains the negative marginal association.

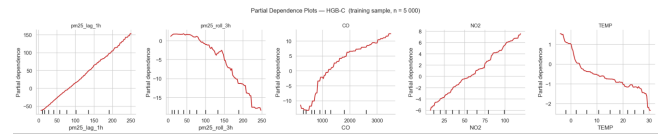


Figure 11. Partial Dependence Plots — HGB-C (training sample, $n = 5,000$).

CO and NO₂ both show positive associations with predicted PM2.5, which is physically intuitive: higher concentrations of these combustion-related pollutants are associated with more polluted conditions, consistent with their shared emission sources.

Finally, temperature shows a weak negative association: higher temperatures are marginally linked to lower predicted PM2.5, reflecting the seasonal pattern identified in the EDA where summer conditions are generally cleaner. However, this effect is small compared to the temporal PM2.5 signal and should be interpreted with caution given the complex interactions between temperature, season, and atmospheric conditions.

Because several predictors are highly correlated, especially `pm25_lag_1h` and `pm25_roll_3h`, these PDPs should be interpreted as descriptive diagnostics of the fitted model rather than independent physical effects.

4.3 Unsupervised Learning Analysis

4.3.1 Preprocessing Recall

The results presented in this section are based on our preprocessing described in Section 3.4. As a reminder, the analysis was conducted at the daily level, with each of the observations representing one monitoring station on one day, obtained by aggregating the raw hourly data to daily means. Prior to clustering, the four most skewed pollutants (PM2.5, PM10, CO, SO₂) were log-transformed (as visible in Figure 12) to reduce the influence of extreme pollution episodes on distance calculations, and temperature and dew point were replaced by their monthly station-level anomalies in order to remove the regular seasonal cycle we identified in the EDA, which is a crucial step to ensure that the clusters reflect genuine air quality regimes rather than simply grouping the days by calendar season. All features were additionally winsorised at the 1st and 99th percentiles, and a `StandardScaler` was applied to bring all nine variables onto a common scale. It is also worth recalling that lag features and rolling averages, which were central to the supervised models, were deliberately excluded here: the goal of the unsupervised analysis is not to predict future pollution levels but to characterise recurring pollution regimes, for which the contemporaneous pollutant and meteorological profile of each day is the relevant information.

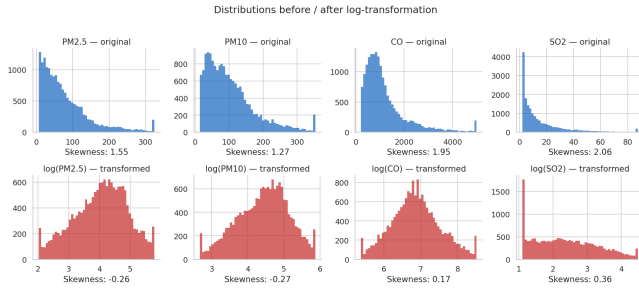


Figure 12. Distributions of PM2.5, PM10, CO and SO₂ before and after the log transformation.

4.3.2 PCA: Variance Structure

Table 2. Variance explained per component.

Component	Variance (%)	Cumulative (%)
PC1	51.2	51.2
PC2	17.0	68.3
PC3	10.2	78.4
PC4	8.1	86.5
PC5	5.0	91.5
PC6	3.9	95.4
PC7	2.1	97.5
PC8	1.9	99.4
PC9	0.6	100.0

Figure 13 presents the PCA scree plot and the cumulative variance explained across the nine principal components. The results reveal a strongly dominant first component, PC1, which alone accounts for 51.2% of the total variance. This is a clear sign that the data is structured around a single dominant axis. PC2 adds a further 17.0%, bringing the two-component coverage to 68.3%, which is the basis for the 2D visualisation presented later. We can see that the cumulative curve then rises more gradually starting from the PC3, reaching 80% by the fourth component. It also seems interesting to state that the 95% threshold is crossed at the sixth PCA. This confirms that while the nine-dimensional feature space contains meaningful information beyond the first two components, the bulk of the structure, and in particular the dominant pollution intensity axis, is already well captured in two dimensions, which can justify the use of PC1 and PC2 as the primary lens for cluster visualisation.

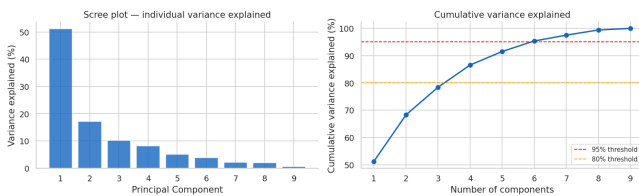


Figure 13. PCA scree plot & cumulative variance explained.

Figure 14 shows the contribution of each variable to PC1 and PC2. PC1 can be interpreted as a pollution intensity axis: PM2.5, CO, PM10, NO₂ and SO₂ all load positively, meaning that days with a high PC1 score are simply days with high concentrations across all combustion-related pollutants. Wind speed loads negatively, which makes intuitive sense as stronger winds tend to disperse particles as we already saw during the EDA. PC2 captures a

different dimension, dominated by temperature and dew point anomalies alongside O₃, and can be thought of as a summer weather axis; it separates days that are unusually warm and humid from the rest, independently of the overall pollution level. These two axes together provide a readable two-dimensional space in which the clusters can be visualised, as we will see further.

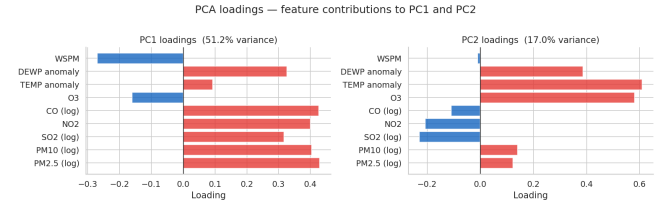


Figure 14. PCA loadings — feature contributions to PC1 and PC2.

4.3.3 Choice of k

Table 3. Inertia and silhouette for 10 first k .

k	Inertia	Silhouette
2	106,201	0.2648
3	86,211	0.2220
4	76,700	0.2057
5	70,202	0.1868
6	65,892	0.1779
7	62,497	0.1713
8	59,062	0.1731
9	56,152	0.1729
10	53,923	0.1613

To select the number of clusters k , we used two diagnostics shown in Figure 15. The elbow plot shows that inertia decreases rapidly from $k = 2$ to $k = 4$, after which the gains become more gradual, suggesting that adding more clusters beyond 4 does not meaningfully improve the compactness of the groups. The silhouette score, which measures how well each observation fits its own cluster compared to neighbouring ones, peaks at $k = 2$ but drops quickly afterwards.

However, by analysing a little bit further, selecting $k = 2$ clusters would only separate ‘polluted’ from ‘clean’ days, which is not very informative. We therefore thought that selecting $k = 4$ was the best compromise. Indeed, it does correspond to the point where the elbow starts flattening a bit, and it produces four groups that are distinct and interpretable enough to tell a meaningful story about Beijing’s pollution patterns, as it will be described in the next section.

It is worth noting, however, that the silhouette scores remain relatively low across all values of k , with a maximum of around 0.26, and is even lower for our chosen $k = 4$, with a value of 0.2057. This suggests that the clusters do not have very sharp boundaries, which is expected given the nature of pollution data: air quality conditions change gradually over time rather than falling into perfectly distinct categories. The clusters should therefore be seen as general tendencies rather than perfectly distinct groups.

The final model chosen has $k = 4$ clusters with a silhou-

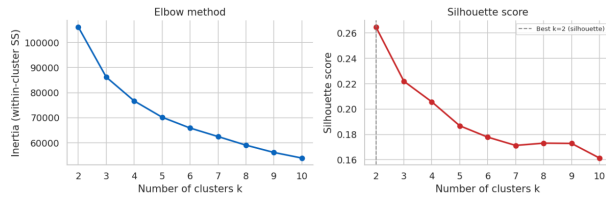


Figure 15. K-Means Elbow method & Silhouette score.

ette score of 0.2057 and the following cluster sizes:

Table 4. Cluster sizes.

Cluster	N	PM2.5	Label
0	3,879	Very high	Severe pollution ep.
1	4,648	High	Summer photo. haze
2	4,254	Low	Clean / windy
3	4,751	Moderate	Moderate conditions

4.3.4 Cluster Profile

Figure 16 shows the mean value of each feature per cluster, allowing us to characterise the four pollution regimes identified by the model. Cluster 0 stands out immediately as the most severely polluted group, with a mean PM2.5 of $166.3 \mu\text{g}/\text{m}^3$, very high CO (2477.3) and NO₂ (88.0), and the lowest wind speed (1.2 m/s). This cluster captures the worst pollution episodes in Beijing, typically associated with winter atmospheric stagnation where pollutants accumulate near the surface with no wind to disperse them. At the opposite end, Cluster 2 is the cleanest group, with a mean PM2.5 of only $19.9 \mu\text{g}/\text{m}^3$ and also the highest wind speed (2.4 m/s), confirming the strong dispersal effect of wind observed in the EDA. Cluster 1 is notable for its high O₃ concentration (101.2) combined with moderate PM2.5 (81.8), which is characteristic of summer photochemical conditions where sunlight drives the formation of ozone rather than direct combustion emissions (Sillman, 1999). Finally, Cluster 3 represents moderate everyday conditions, with intermediate values across all pollutants and no particularly extreme feature, making it the most ‘average’ group of the four. Together, these four profiles paint a coherent picture of Beijing’s main air quality regimes, ranging from clean and windy days to severe winter pollution episodes.

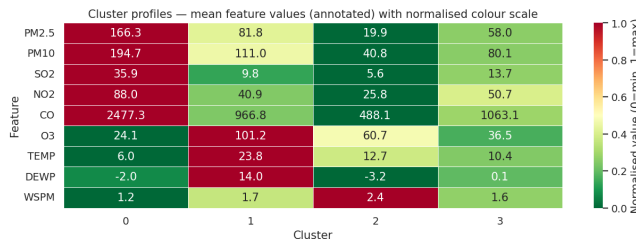


Figure 16. Cluster profiles — mean feature values (normalised colour scale).

Figure 17 shows the seasonal composition of each cluster, and provides an important validation of our preprocessing choices. Note that we have already attributed each name to the different clusters based on their profiles. Cluster 0 is labelled ‘Severe pollution episodes’, Cluster 1 ‘Summer photochemical haze’, Cluster 2 ‘Clean / windy’, and Cluster 3 ‘Moderate conditions’.

We can see that the Clean/windy cluster is the most balanced of the four, with days spread relatively evenly across all four seasons, confirming that clean air conditions are not exclusive to any particular time of year but rather driven by meteorological factors such as wind speed. The Moderate conditions cluster follows a similar pattern, though with a slightly higher share of autumn days. The Severe pollution episodes cluster is heavily dominated by winter (46.9%) and autumn (29.1%), with almost no summer days (2.4%), which aligns perfectly with the known mechanisms of winter pollution in Beijing, which is characterized by cold temperatures, heating emissions, and atmospheric stagnation (An et al., 2019). Finally, the Summer photochemical haze cluster is, as expected, almost entirely composed of summer (54%) and spring (31%) days, with very few autumn and winter observations. This seasonal distribution confirms that our clusters are not simply grouping days by season, which was the main risk identified in the preprocessing stage, but are instead capturing genuinely distinct pollution regimes that happen to correlate with seasons for physical reasons.

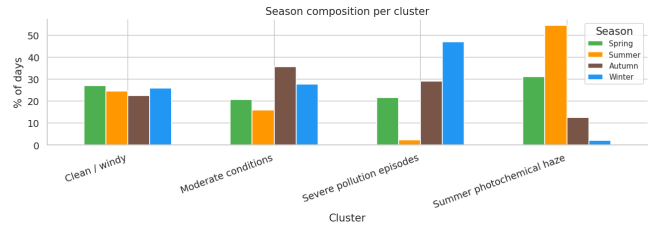


Figure 17. Season composition per cluster.

Figure 18 shows the distribution of daily mean PM2.5 values within each cluster, and confirms that the four groups are well separated in terms of actual pollution levels. Logically, the severe pollution episodes cluster has by far the highest concentrations, with a median around $150 \mu\text{g}/\text{m}^3$ and a wide interquartile range extending up to $220 \mu\text{g}/\text{m}^3$. The Summer photochemical haze cluster sits in an intermediate range with a median around $75 \mu\text{g}/\text{m}^3$, while the Clean/windy cluster is clearly the lowest, with a median below $20 \mu\text{g}/\text{m}^3$ and a very narrow distribution indicating consistently clean conditions. The Moderate conditions cluster, as expected, falls between the two extremes with a median around $55 \mu\text{g}/\text{m}^3$. It is also worth noting that all four clusters contain some outliers, which is expected given the nature of pollution data, but the overall separation between groups is clear enough to validate the relevance of the $k = 4$ solution.

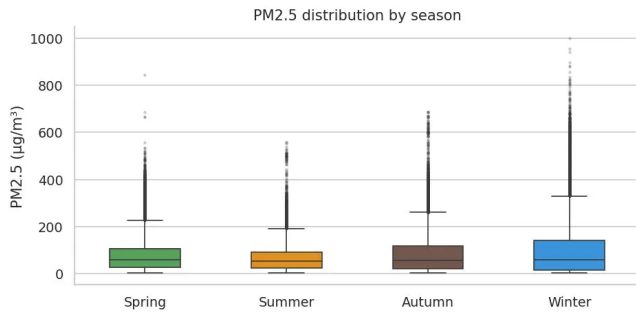


Figure 18. PM2.5 distribution by cluster.

4.3.5 PCA 2D Visualisation of Clusters

Figure 19 provides a two-dimensional visualisation of the clustering results projected onto the first two principal components. As a reminder, the clustering was performed in the full nine-dimensional space, the PCA projection is used here purely to allow a visual representation of the results. The left plot shows the four clusters in PCA space, and the separation is fairly visible: the Clean/windy cluster (green) sits on the left side of the plot, corresponding to low values on PC1, while the Severe pollution episodes cluster (blue) occupies the right side, confirming that PC1 effectively captures pollution intensity. The Summer photochemical haze cluster (orange) sits in the upper part of the plot, separated from the others along PC2, which reflects its distinct meteorological profile. The right plot, coloured by actual PM2.5 levels, further validates this structure: PM2.5 concentrations increase consistently from left to right along PC1, which is exactly what we would expect given the loadings discussed earlier. We can note that the overlap between clusters in the centre of the plot is consistent with the low silhouette scores observed, and simply reflects the fact that pollution levels change gradually over time rather than jumping from one regime to another. The $k = 4$ solution should therefore be interpreted as a useful descriptive framework rather than a strict classification.

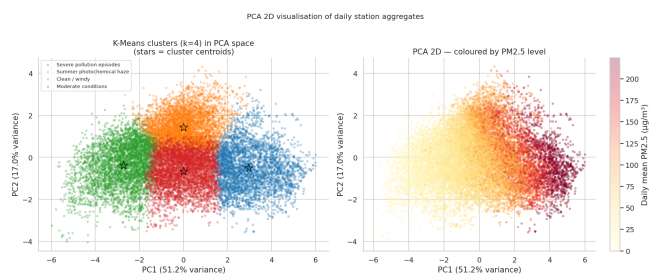


Figure 19. PCA 2D visualisation of daily station aggregates.

5 Conclusion

This study investigated whether machine learning models can effectively predict air pollution levels and identify underlying patterns in Beijing’s air quality data. Using the Beijing Multi-Site Air Quality Dataset, we applied both supervised and unsupervised learning techniques to better understand the key drivers of PM2.5 concentrations across 12 monitoring stations over a four-year period.

The exploratory data analysis revealed several important

patterns: a strong seasonal cycle with winter being by far the most polluted season, a clear daily cycle driven by atmospheric physics, and a dominant role of wind speed as a natural dispersal mechanism. These findings directly informed the modelling choices made in both parts of the study.

On the supervised learning side, three model families — OLS, Ridge regression, and Histogram Gradient Boosting — were evaluated across three feature sets of increasing richness. The most important finding was that feature engineering mattered far more than model complexity: using PM2.5 lag and rolling features reduced the test RMSE by more than 70% compared to using weather and temporal variables alone. The final model, HGB with feature set C, achieved a test RMSE of $17.04 \mu\text{g}/\text{m}^3$ and a test R^2 of 0.958, explaining approximately 96% of the variance in held-out hourly PM2.5 concentrations. Feature importance analysis confirmed that the 1-hour PM2.5 lag is by far the dominant predictor, followed by co-pollutants such as CO and NO₂. While this model performs very well for short-term prediction, it is important to note that its strong performance relies heavily on the availability of the most recent PM2.5 observation, which limits its usefulness for longer-horizon forecasting.

On the unsupervised side, the K-Means clustering identified four distinct pollution regimes: Clean/windy days, Moderate conditions, Summer photochemical haze, and Severe pollution episodes. These four profiles paint a coherent and physically interpretable picture of Beijing’s main air quality regimes. The PCA analysis confirmed that pollution intensity, driven by combustion-related pollutants, is the dominant axis of variability in the data, while a secondary axis captures abnormal meteorological warmth associated with summer conditions.

Overall, both the supervised and unsupervised approaches converged on the same key findings: recent PM2.5 history, wind speed, CO and NO₂ are the most important variables driving air quality variations in Beijing, and seasonal and meteorological conditions play a central role in shaping pollution levels. The convergence of these two independent approaches reinforces the robustness of the conclusions.

5.1 Limitations

Several limitations of this study should be acknowledged.

First, the dataset covers the period from March 2013 to February 2017, which means it is now almost a decade old. Beijing has undergone significant policy changes and industrial restructuring since then, including major air quality improvement plans implemented after 2017. The pollution patterns identified in this study may therefore not fully reflect the current situation in Beijing, and the predictive models trained on this data may not generalise well to more recent conditions.

Second, the dataset contained a non-negligible number of missing values, which had to be handled prior to modelling. We addressed this through a temporal inter-

polation strategy with a maximum gap of 6 consecutive hours, and station-level median imputation for larger gaps. While we believe this approach is well-suited to the continuous nature of pollution data, it inevitably introduces some approximation in the dataset, and the imputed values remain estimates rather than true observations.

Third, the supervised model's strong performance is largely dependent on the availability of the most recent PM_{2.5} observation as an input feature. While this is realistic in a monitoring context, it means the model is essentially performing one-step-ahead prediction rather than genuine multi-horizon forecasting, which limits its practical applicability for early warning systems or longer-term planning.

Fourth, regarding the unsupervised analysis, the silhouette score of our final $k = 4$ clustering solution reached only 0.206, which is below the commonly accepted threshold for a well-separated clustering structure. This reflects the inherently continuous nature of air quality data, where pollution conditions transition gradually rather than falling into perfectly distinct categories. While the four clusters are physically interpretable and consistent with domain knowledge, they should be understood as a descriptive framework rather than a strict classification. It is also worth noting that silhouette scores in the range of 0.20–0.30 are commonly reported in the environmental data clustering literature, which suggests our result is not unusual for this type of data.

5.2 Outlook

Several directions could build on this work. Incorporating more recent data would allow the models to reflect Beijing's current air quality situation and the impact of recent policy interventions. Extending the analysis to other Chinese cities or to a global dataset would help assess the generalisability of the patterns identified. On the modelling side, deep learning approaches such as LSTMs could better capture the long-range temporal dependencies in pollution data and potentially improve performance for longer-horizon forecasting. Finally, integrating additional variables such as traffic counts, industrial output, or satellite imagery could improve both the predictive performance of the supervised models and the interpretability of the clustering results.

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